

Natural gas, trade openness and economic growth in Peru: 1965–2022

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ABSTRACT

Natural gas is a major fuel in Peru, accounting for nearly 45 % of all energy sources used to generate electricity. This research aims to analyze the dynamic association between economic growth and natural gas consumption (NGC), considering Peru's trade openness (Tr) from 1965 to 2022. The augmented Dickey–Fuller unit root test is utilized to test the stationary properties of the log series of real gross domestic product (GDP), NGC, and real Tr. We used the autoregressive distributed lag (ARDL) cointegration approach to determine the long-run (LR) equilibrium association between the variables. As the second cointegration method, we used the Engle–Granger test. The results of the ARDL test support the cointegration of all variables, we find the opposite case with Engle–Granger method. The Granger causality results support the neutrality hypothesis between GDP and NGC, a two-way causality association between Tr and GDP, and the neutrality hypothesis between NGC and Tr. Nevertheless, in the LR, cointegration confirms the growth hypothesis between GDP and NGC. Similarly, we find that in the LR, the first lag of Tr positively impacts the current GDP value. Meanwhile, in the short run, Tr has a positive effect on Peru's GDP. Then, using the findings of cointegration among all variables, we can conclude that NGC and Tr are the drivers of economic growth in Peru in the LR. Finally, we discuss the policy implications of our results.

1. Introduction

Since the beginning of the 21st century, natural gas (NG) has played an important protagonist in energy transition (ET) toward more sustainable energy sources (ESs) [1]. NG is an energy resource (ER) that, due to its characteristics -it is the least polluting fossil fuel (PFF)

compared to oil and carbon, is abundant in nature, and has low costs in relation to renewable energies-has been classified as a “ET fuel,” i.e., as a bridge in the global transition from EM based on the use of PFF to matrices based on nonpolluting renewable sources (RS) [2–5]. Likewise, some academics argue that NG can be used with renewable energy (RE) to help mitigate intermittency issues, such as the fact that renewable

Abbreviations: ADF, Augmented Dickey–Fuller; AIC, Akaike information criteria; ARDL, Autoregressive distributed lag; CO₂, Carbon dioxide; BCRP, Central Reserve Bank of Peru; CH, Conservation hypothesis; CNG, Compressed natural gas; CT, Cointegration test; DF-GLS, Dickey–Fuller generalized least squares; DNG, Dry natural gas; ELT, Economic growth-led trade; EC, Energy consumption; EI, Energy Institute; ECM, Error correction model; EJ, Exajoules; EG, Economic growth; EPG, Electricity power generation; EM, Energy matrix; ER, Energy resource; ES, Energy source; ESe, Energy sector; ESu, Energy supply; ET, Energy transition; FH, Feedback hypothesis; I(1), First-order integrated processes; GC, Granger causality; GDP, Gross domestic product; GFCF, Gross fixed capital formation; GH, Growth hypothesis; HQ, Hannan–Quinn; HTB, Hydrocarbon trade balance; IAA, Innovative accounting approach; IT, International trade; JB, Jarque–Bera; LF, Labor force; LNG, Liquefied natural gas; LPG, Liquefied petroleum gas; MMcf, Million cubic feet per day; Mtpa, Million tons per annum; NG, Natural gas; NGC, Natural gas consumption; NGL, Natural gas liquids; NH, Neutrality hypothesis; H₀, Null hypothesis; PC, Per capita; PHS, Peruvian hydrocarbons Society; PEN, Peruvian Nuevo sol; PFF, Polluting fossil fuels; PE, Primary energy; RE, Renewable energies; RS, Renewable sources; SD, Supreme decree; SDe, Sustainable development; SDGs, Sustainable development goals; SIC, Schwarz information criteria; S&LR, Short- and long-run; SR, Short-run; Tcm, Trillion cubic meter; TLE, Trade-led economic growth; Tr, Trade openness; TY, Toda–Yamamoto; URT, Unit root test; VAR, Vector autoregression; VECM, Vector error correction model; VNG, Vehicle natural gas; I(0), Zero-order integrated processes.

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wind and solar sources are not always available when they are required [2,3,6–8]. However, given that NG releases CO₂ into the environment, there is literature arguing that investments in infrastructure with a long useful life are destined to develop NG ET to zero or near zero carbon systems [4,6,9,10].

Based on [11], the exploitation of NG in Peru from 2004 to 2019 had an average annual direct contribution to the Peruvian economy was close to 1.1 % of GDP. In terms of Peruvian NG exports, the value was 3166.2 (US\$MM FOB) in 2023 [12], it represents 2.3 % of total Peruvian exports by economic activity group in 2023. Thus, the Tr must be incorporated in the NGC-EG nexus [13]. The NG industry has also impacted the industrial, commercial, residential, electrical, and transportation sectors [1]. In Peru, since the commercial operation of NG's Camisea project began in August 2004, the NG has made it possible to transform and expand the Peruvian energy matrix (EM) toward a less polluting ER, providing a more efficient and economical ES than coal or diesel in electricity power generation (EPG) through thermoelectric plants, having to NG as a substitute fuel for liquefied petroleum gas (LPG) in the commercial sector; as an energetic alternative for oil, LPG, and coal in the industrial sector; as an alternate ER for LPG in the urban residential sector; as a substitute ES for firewood and biomass in the rural residential sector; and as an alternative for gasoline or diesel in the vehicle sector.

With the establishment of the Ministry of the Environment and the promulgation of DL N° 1002, the year 2008 marked a watershed moment in Peruvian environmental promotion. Both measures aim to enhance the population's quality of life while also protecting the environment by promoting the use of renewable ERs. The investment in EPG will drive this improvement [14].

In 2010, Supreme decree (SD) N° 064-2010-EM approved the National Energy Policy of Peru 2010–2040, with nine objectives: (1) achieve a diversified EM, prioritizing RS and energy efficiency; (2) promote competitive energy supply (ESu); (3) facilitate unrestricted access to ESu; (4) optimize energy production and consumption to the maximum extent possible; (5) achieve energy independence; (6) promote an environmentally responsible energy sector (Ese) with low CO₂ emissions, incorporating the principles of SDe; (7) promote the development of NG and its application in different sectors such as industry, households, commerce, and transport, with efficient pipelines that optimize distribution; (8) strengthen the institutions of the ESe; and (9) collaborate with the energy industries of the region to make the LR vision a reality [15].

Within the framework of public policies for energy transition, the Peruvian State has implemented a set of regulations to promote the transition towards a diversified and sustainable energy matrix, with an emphasis on the integration of non-conventional RE into the electricity grid. These measures aim to fulfill the commitments of the 2015 Paris Agreement, ratified by SD N° 058-2016-RE [16]. Law N° 30754 of 2018 establishes the creation of guarantee funds to stimulate investment in renewable and clean energies [17]. In 2022, SD N° 003-2022-MINAM declared the climate emergency as a matter of national interest, setting as goals a 20 % share of non-conventional RE in the energy matrix and a reduction of 40 % in greenhouse gas emissions by 2030 [18]. However, in parallel, regulations have been implemented for the massification of NG, such as SD N° 010-2016-EM [19] and SD N° 037-2018-EM [20].

According to Ref. [21], the association among energy consumption (EC) and EG is a highly important topic in many countries, as it allows us to determine the link between the constraints of the energy sector and economic performance, with particular attention to CO₂ emission control.

Agreeing to the previously described context, it is imperative for policymakers to understand the dynamics of the causal association between NG consumption (NGC) and economic growth (EG). This understanding will enable the design of robust, solid, and coherent policies that are consistent with each other and with the objectives set in

environmental, economic, and energy matters.

NG presents itself as a viable alternative to oil scarcity, being a non-renewable energy source that, compared to other fossil fuels, generates lower CO₂ emissions. In this sense, NG is used as an input in electricity production and plays a crucial role as a raw material for the industrial sector in developed, emerging, and developing economies. This sector plays a key role in the economic activity of countries, whose development drives EG Ref. [22]. Four hypotheses are theoretically and empirically tested in the NGC-EG nexus study. Growth hypothesis (GH), feedback hypothesis (FH), conservation hypothesis (CH), and neutrality hypothesis (NH). GH claims a positive effect or one-way causality from NGC to EG Refs. [22–31]. The CH states that there is a positive effect or unidirectional causality from EG to NGC [24,26,32–34]. The FH establishes that both the growth and CHs are validated concurrently [13,21,24,25,27,29,30,32,35,36]. NH hypothesizes that EG does not cause or impact NGC, and vice versa [24,29]. Numerous studies have supported these four hypotheses, either through causality or cointegration (through significant positive impacts¹ either in the SR or in the LR, or even in both time horizons).

The NGC emerges as an effective measure to improve energy efficiency and contribute to pollution reduction, both in international trade activities (exports and imports) and in the transportation systems associated with these activities [37]. In the study of the Tr-NGC relationship [38,39], have found empirical evidence of a unidirectional causal relationship from NGC to Tr in OECD countries (1991–2013) and in GCC countries (1980–2012), respectively. [27] reported findings in favor of a unidirectional causal relationship from Tr to NGC in Malaysia for the period 1971–2012. [37] found support for a bidirectional causal relationship between Tr and NGC in fifteen developing countries with the highest consumption of NG for the period 1990–2012. Finally, [22,40] have found empirical evidence of a neutral causality between Tr and NGC in Saudi Arabia (1968–2016) and in selected sub-Saharan African countries (1998–2017), correspondingly.

According to the tenets of neoclassical theory, international trade benefits nations by providing access to broader markets, enabling the exchange of goods and services, and thereby expanding trade prospects. Additionally, it encourages countries to specialize in the production of goods and services where they are most efficient while importing those where they are less efficient. This specialization leads to improved production and more efficient resource utilization, fostering EG Ref. [41]. In the dynamic relationship between Tr and EG, evidence supporting the trade-led economic growth (TLE) hypothesis is reported by Refs. [27,40,42] in Malaysia (1971–2012), in selected sub-Saharan African countries (1998–2017), and in Cote d'Ivoire (1965–2014), respectively. In research papers by Refs. [43–45], the economic growth-led trade (ELT) hypothesis is supported in India, China and South Africa (1981–2013), in Pakistan (1971–2011), and in Turkey (1950–2014), correspondingly. [38,39,46–48] found a two-way causal relationship between Tr and EG in OECD countries (1991–2013), GCC countries (1980–2012), Asia-Pacific countries (1980–2017), Australia (1960–2012), and in forty-six emerging market and developing economies (1990–2014), respectively. Finally, [22,33,43] found findings in favor of a neutral causal relationship between Tr and EG in Saudi Arabia (1968–2016), Japan (1980–2020), and in Brazil (1981–2013) and Russia (1989–2013), correspondingly. While the literature on the

¹ It is important to highlight that both causality and negative impacts between NGC and EG are also reported in the empirical literature. [32] found that for Malaysia during the 1971–2012 period, GDP had a negative and significant effect on NGC, both in the S&LR. [35], using an asymmetric nonlinear causality test (Hatemi-J) —which distinguishes the causal effect of positive variations from negative variations— found a one-way causality from EG to the negative impact of NGC in Nigeria for the period 1981–2015. [36], employing the ARDL cointegration approach, found that for Turkey during the 1995–2012 period, NGC negatively impacted GDP, in S&LR.

NGC-EG nexus is extensive, there is a lack of research analyzing its dynamic impact in the case of Peru, no research has analysed the effect of NG consumption (NGC) and trade openness (Tr) on economic growth (EG) in this context.

This study aims to fill this gap in the literature by making three main contributions: (1) broaden the empirical literature on the NGC-EG nexus; (2) provides empirical evidence that allows determining the causal direction between the economic variables analyzed, with the aim of providing guidelines to public policy designers for the creation and implementation of measures that guarantee SDE; and (3) examine whether Tr's contribution to EG is direct or indirect.

Therefore, the main objective of this article is to analyze the dynamics of the NGC-EG nexus in Peru, an upper-middle-income country, with Tr as a non-omitted variable. Furthermore, the following three secondary objectives are established: (i) empirically verify which of the four hypotheses about the NGC-EG relationship are verified in Peru; (ii) empirically confirm the type of causal nexus between Tr and gross domestic product (GDP) that is validated in Peru; and (iii) empirically test the type of causal nexus between Tr and NGC that is verified in Peru. Using annual series of GDP, NGC, and Tr. Our research is related to the United Nations Sustainable Development Goals (SDGs): NGC to SDG 7, real GDP to SDG 8, and Tr to SDG 12. We analyze stationarity, the Granger causality (GC), and cointegration between all series in Peru from 1965 to 2022.

The remaining sections of the paper are organized as follows. The results of previous research are summarized in Section 2. Section 3 presents a brief overview of the global EM and Peruvian EM. Similarly, this section briefly overviews Peru's NG industry and its major impacts. Section 4 discusses the materials, variables, data, and methodology used. Section 5 make available the obtained results. Section 6 examines the obtained results. Section 7 outlines the conclusions and policy recommendations.

2. Literature review

The following sections summarize an empirical literature review of the NGC-EG relationship from 2013 to 2023. To facilitate the comparison of results from our research, based on time series data, with other research, our review only includes studies that used time series data.²

2.1. Growth hypothesis

The following articles found evidence for a GH between NGC and EG through causality or cointegration. [23] empirically analyze the relationship between NGC and EG in Pakistan from 1972 to 2010, controlled by X PC, Capital Use per capita (PC), and LF PC. The paper uses URTs (ADF, PP, Dickey-Fuller generalized least squares [DF-GLS]) and CTs (ARDL bounds and Johansen). Academics validated the GH using the innovative accounting approach (IAA) GC. Cointegration confirmed GH in Pakistan in S&LR. As a first finding, [24] estimated the dynamic relationship between NGC and real GDP in nine Asian countries between 1980 and 2012, controlling for X and gross fixed capital formation (GFCF). The author found favorable evidence for GH in China by means of ARDL cointegration and VECM GC approaches. With Toda-Yamamoto's (TY) GC, this author validated the GH in Thailand. As a first outcome [25], using log series of real GDP PC, NGC PC, exports PC, real GFCF, and labor force (LF) participation rate in France from 1990 to 2014, found favorable evidence for GH using the ARDL bounds cointegration approach, both in the short- and long-run (S&LR). As a first result, [26] validated the unidirectional causal nexus between NGC and real GDP in Malaysia between 1980 and 2014 using the ARDL and Bayer & Hanck cointegration approaches, both in S&LR, with GFCF, the KOF

index of globalization, and CO₂ as control variables. As a first result, [27] applied ARDL and Bayer & Hanck CTs to the log series of the real GDP PC, NGC PC, GFCF, Tr PC, and FDI PC, confirming the GH by cointegration between NGC and EG in Malaysia from 1980 to 2010, in S&LR. The GH was also verified using GC in the SR. [22], examine the NGC-real GDP nexus in Saudi Arabia from 1968 to 2016. The outcomes of the ARDL cointegration method and the TY GC supported the GH among the research variables. [28], examines the NGC-EG nexus using Malaysian data from 1982 to 2020 and the ARDL cointegration method and the error correction model (ECM). The cointegration of industrial GDP, industrial NGC, FDI inflows, LF, GFCF, Tr, and FD is verified using GDP as the dependent variable. The cointegration of S&LR supports the GH between NGC and EG. As a first finding, using unit root test (URTs); augmented Dickey-Fuller [ADF], (PP)-without and with breakpoint and Hacker & Hatemi-J causality to the log series of the real GDP PC and NGC PC [29], reported evidence of GH among NGC and EG in Turkey -supported by Hatemi-J causality- for the periods 1996-2010 and 1997-2011. As a first result, [30] tested the cointegration between GDP, LF, NGC, and GFCF in Indonesia from 1980 to 2017 by means of the ARDL bounds test and the estimation of the corresponding ECM, concluding that cointegration verifies the GH in S&LR. [31], using Johansen's cointegration test (CT), found findings in favor of GH among NGC and GDP PC in Pakistan from 1977 to 2013, in LR, controlled by LF and GFCF. Similarly, these authors verified the GH using the results of the VECM GC approach.

2.2. Conservation hypothesis

Some studies that found evidence in favor of GH among NGC and EG using various causality approaches include the following. As a second result, [24] used the TY GC test and found evidence of a CH among NGC and EG in Bangladesh, Singapore, and Malaysia. As a final result, [26] analyze the dynamic relationship between NGC and EG in Malaysia between 1980 and 2014. The outcomes of the GC test in this study validate the CH. As a first result, [32] applied URTs (ADF and PP), CTs (Bayer & Hanck, Pesaran-Shin-Smith and Johansen), VECM GC, and innovative accounting approach (IAA) GC, to real GDP, NGC PC, real exports PC, real capital stock PC and LF PC in Malaysia during the period 1971-2012, and found a cointegrated log series and evidence of a unidirectional GC relationship from EG to NGC, in the SR. [33] investigate the NGC-EG nexus in Japan, controlled by CO₂, Tr, and GFCF, from 1980 to 2020. The results of the TY GC validated the CH. Using the VECM GC approach, [34] found favorable evidence of CH among NGC and EG in Pakistan between 1980 and 2010.

2.3. Feedback hypothesis

The following studies found findings in favor of FH between NGC and EG using some type of causality method. [21] used machine learning experiments in Germany and Japan from 1970 to 2018 to analyze the causal nexus between NGC and the PC GDP Dependency (D2C) algorithm. The results obtained for these academics confirmed the FH between HGC and EG in both Germany and Japan. As a third outcome, [24] used the TY GC approach to confirm FH between NGC and EG in Japan, India, and Pakistan between 1980 and 2012. As a second finding, applying a GC VECM to the first difference of the log series of real GDP, NGC, exports, real GFCF, and the participation rate in LF, [25] found support for FH between NGC and GDP in France (1990-2014), both in the S&LR. As a final result, [27] verified the FH between NGC and EG in Malaysia between 1980 and 2010, in the LR. As a second finding, [29] support the FH between the real GDP PC and the NGC PC in Turkey during the sub-period 2001-2014. As a final result, [30] validated the bidirectional causal nexus between NGC, GDP, LF, and GFCF by using the variance decomposition method and confirming the FH between NGC and GDP in Indonesia from 1980 to 2017. As a last finding, [32] found evidence of a bidirectional GC relationship in Malaysia from 1971

² Those readers interested in papers using panel data may refer, for example, to Refs. [49-53].

to 2012, in the SR. Using log series of real GDP, NGC, LF, and capital, [35] found support in favor of a two-way causality between the positive impact of NGC and GDP –verified by asymmetric nonlinear Hatemi-J causality test– in Nigeria from 1981 to 2015. Meanwhile, [36] found a bidirectional GC relationship between NGC and GDP using the VECM GC approach for Turkey, controlling for GFCF and LF for the 1995–2012 period in both S&LR. [13] analyzed the NGC–EG nexus in Turkey from 1988 to 2021. To accomplish this, they used log series of real GDP, NGC, CO₂, and the number of technological patents. The methodology included analyzing the series' stationarity using the ADF and PP URTs, testing for cointegration between variables using the Johansen test, and analyzing causality using the Markov–Switching vectorial autoregressive (VAR), Markov–Switching GC, Fourier VAR, and Fourier GC. The outcomes show that all of the log series are stationary in first differences and that there is no cointegration among all variables. Finally, Fourier GC verified a feedback nexus between NGC and GDP in Turkey.

2.4. Neutrality hypothesis

The following articles found findings in favor of NH between NGC and EG using some type of causality test. Using the TY GC test, [24] found evidence of an NH among NGC and EG in South Korea from 1980 to 2012. As a final finding, [29] found no causal relationship between NGC and EG in Turkey from 1983 to 2017.

3. Overview of world energy matrix and the Peruvian energy matrix: the natural gas industry in Peru and its main impacts

3.1. An overview of world energy matrix

Primary energy (PE) consumption in the world amounted to 597.41 EJ (EJ) in 2021, whereas in 2022, the world consumption was 604.04 EJ. The PE matrix consumed by the types of fuels in the world comprised oil, hydroelectric generation, NG, renewables, coal, and nuclear energy. From 2021 to 2022, the share of oil in this matrix increased from 30.9 % (184.86 EJ) to 31.6 % (190.69 EJ), the share of hydroelectric generation decreased from 6.8 % (40.40 EJ) to 6.7 % (40.68 EJ), the percentage of NG decreased from 24.5 % (146.41 EJ) to 23.5 % (141.89 EJ), the fraction of renewables increased from 6.7 % (39.97 EJ) to 7.5 % (45.18 EJ), the share of coal decreased from 26.9 % (160.43 EJ) to 26.7 % (161.47 EJ), and the share of nuclear energy decreased from 4.2 % (25.33) to 4.0 % (24.13) [54]. Therefore, in 2022, NG became the third ER in the primary EM of the planet, only surpassed by oil and coal.

In 2021, the global EPG was equal to an impressive amount of 28520.2 TWh. In this year, the EM of the electricity produced on the planet was composed of hydro 15.0 % (4288.78 TWh), NG 23.0 % (6565.55 TWh), nuclear energy 9.8 % (2802.55 TWh), renewables 12.8 % (3664.58 TWh), coal 35.8 % (10211.06 TWh), oil 2.6 % (733.48 TWh), and other 0.9 % (254.17 TWh).³ In 2022, the EPG was 29165.1 TWh. In this year, worldwide electrical generation by type of ES was composed of hydro 14.9 % (4334.19 TWh), NG 22.7 % (6631.40 TWh), nuclear energy 9.2 % (2679.01), renewables 14.4 % (4204.31 TWh), 35.4 % coal (10317.16 TWh), 2.5 % oil (728.60 TWh), and other 0.9 % (270.46 TWh) [54]. Consequently, in 2022, NG was the world's second most important ES in EPG, below coal and above hydroelectric generation. Furthermore, between 2021 and 2022, the percentage change in the production of NG in the worldwide EPG was 1.0 %, the percentage variation in the production of RS was equal to a notable 14.7 %, while nuclear energy showed the greatest decline (4.4 %).

³ Takes in account unclassified production, statistical discrepancies and sources not elsewhere specified –hydraulic pumping, non-renewable waste and heat from chemical sources– [54].

3.2. A brief overview of the energy matrix and natural gas industry in Peru

3.2.1. Peruvian energy matrix

To analyze Peruvian ET, it is essential to know the composition of the primary EM and the main ER consumed. PE consumption in Peru was equal to 1.14 EJ in 2021, while in 2022 Peruvian consumption was 1.21 EJ. In Peru, primary EM comprises oil, hydroelectric generation, NG, renewables and coal. From 2021 to 2022, the share of oil in this matrix decreased from 41.9 % (0.45 EJ) to 41.2 % (0.50 EJ), the share of hydroelectric power decreased from 26.3 % (0.30 EJ) to 23.2 % (0.28 EJ), the percentage of NG increased from 25.3 % (0.29 EJ) to 28.7 % (0.35 EJ), the fraction of renewables decreased from 4.6 % (0.052 EJ) to 4.5 % (0.054 EJ), and the share of coal increased from 1.9 % (0.02 EJ) to 2.4 % (0.03 EJ) [54]. From the above it follows that, in 2022, NG became the second ER in the Peruvian primary EM, only surpassed by oil and closely followed by hydroelectric power generation. Similarly, with respect to RE, its share was less than 5 % of Peru's PE consumption in both 2021 (4.6 %) and 2022 (4.5 %).

3.2.2. Supply in the domestic natural gas market

Peru's NG supply chain consists of six activities: exploration, exploitation (production), processing, transportation, distribution, and commercialization. Exploration and extraction activities are considered competitive. Although the regulatory authority does not regulate the extraction segment - OSINERMIN - only for lot 88 of Camisea in Cuzco, the maximum prices are established at the wellhead following the conditions established by the Peruvian State in the respective license contracts; prices in the remaining wells are set in accordance with international market prices. Similarly, because of the high levels of infrastructure investment and the subadditivity of costs in the pipeline network of transport and distribution activities, these activities are considered natural monopolies and are therefore subject to rate regulation by OSINERMIN. However, in the transport and distribution of compressed NG (CNG) and liquefied NG (LNG) via tank trucks, known as "virtual" transport, for residential, industrial, commercial, or vehicular purposes, prices are determined in a competitive environment in which supply and demand interact without state intervention. Finally, commercialization activities develop based on free competition and free access to economic activity [55].

Exploration activities allow for the discovery of new NG deposits, increasing the country's supply of this valuable resource and encouraging investment in the NG industry. Peru's proved NG reserves were 0.276 (Tcm) in 2019 and 0.261 (Tcm) in 2020, corresponding to a 0.1 % share of total global reserves in 2020 (188.1 Tcm) [54]. According to Promigas Peru (2023), Peru's autonomy index of proved reserves was 18.1 at the end of 2022, indicating that if annual NG production in 2022 remains constant and no new proved reserves are added, Peru will have a guaranteed NG supply for the next 18 years.

Prior to the Camisea Project's implementation and operation (in August 2004), NG production in Peru was concentrated in Aguaytía (lot 31C in Ucayali) as well as on the coastal and in the northern (Piura and Tumbes) part of the country. From 2004 to the present, NG production has been concentrated in Camisea lots 56, 57, and 88; Plus Petrol Peru Corporation operates lots 56 and 88 in Cuzco, while lot 57, located between Junín, Ucayali, and Cuzco, is operated by Repsol Exploration Peru [1]. The audited production of NG in 2022 was 1101 million cubic feet per day (MMcfd), while in 2023, it was nearly 1327 MMcfd [56]. The audited production of NG in Peru of lots 56, 57, and 88 of Camisea in 2022 and 2023 represents slightly more than 95 % of total audited production in Peru, reflecting the Camisea NG fields' dominant role in the SDe of Peru's energy industry and economy. In 2004, SD N° 032-2004-EM approved regulating hydrocarbon exploration and production activities [57].

After being extracted from the Camisea gas fields in Cuzco (lots 56, 57, and 88), the NG is transported through the collection and injection

systems operated by Pluspetrol Peru Corporation (lots 56, 57, and 88) and Repsol Exploration Peru (lots 56 and 57) to the Malvinas gas separation plant in Cuzco for processing. This plant processes NG to produce dry NG (DNG) and NG liquids (NGL). Similarly, after NG is extracted from the Aguaytía gas field in Ucayali (lot 31C), it is transported by Energy Peru Aguaytía's collection and injection systems for processing at the Curimaná gas separation plant, which is located within the Aguaytía gas field. This plant processes NG to produce DNG and NGL.

To meet domestic demand, NG can be transported via high pressure gas ducts (for DNG and NGL) or by land (for CNG and LNG). To meet foreign demand for NG, LNG is transported by sea. The concessionaire company Peruvian gas transporter (PGT) transports DNG from the Malvinas plant to Lima (city gate in Lurín), Ayacucho, and Ica to meet demand from various users, including thermoelectric generators, vehicles, homes, businesses, and industries.⁴ Likewise, DNG can be transported from the Malvinas plant to the Chiquintirca compressor plant in Ayacucho via a gas duct system operated by PGT. DNG is transported from this last plant to the Peru LNG liquefaction plant in Pampa Melchorita (Cañete) via the company's gas pipeline. NGL can be transported from the Malvinas plant via the PGT-operated polyduct to the Pluspetrol fractionation plant in Pisco, where it will be converted into gasolines, diesel 2, and LPG before being delivered to their respective markets via trucks and ships. Furthermore, the Energy Peru Aguaytía collection and injection system can transport NG from lot 31C in Ucayali to the Curimaná separation plant. The DNG from this plant is transported to the Aguaytía thermal power plant and the Yarinacocha fractionation plant, while the NGL produced at the Curimaná plant are transported to the Yarinacocha plant for conversion into LPG and gasoline. Enel Generation Piura also transports DNG from the Pariñas Gas Plant to the Malacas Thermal Power Plant and the Talara Refinery in the northwest of Peru's coast [1]. According to Ref. [56], the cumulative capacity of Peru's NG transportation infrastructure in 2022 was 1568 MMcfd, and this infrastructure has been maintained for several years (since 2016) with little development [1]. Similarly, the transported volume of NG was 994 MMcfd in 2021, but increased to 1229 MMcfd in 2022, representing a 24 % increase between 2021 and 2022 [56]. In 2004, Peru promulgated SD N° 018-2004-EM, approving regulations for NG transportation through ducts [58]. In 2007, SD N° 081-2007-EM was promulgated, which approves the regulation of hydrocarbon transportation through ducts [59].

Distribution activities include supplying NG to final users (regulated or independent clients) via low- and medium-pressure gas duct networks. Peru currently has four distribution concessions: Gas natural de Lima and Callao S.A., Contugas SAC, Gases del Pacífico SAC, Gases del Norte del Perú SAC [60]. However, it is important to note that between December 2017 and December 2020, Naturgy Peru S.A. was in charge of the south west concession. When the concession expired through SD N° 009-2020-EM, the state company PETROPERÚ temporarily assumed administration (between December 2020 and December 2023) [61]. PETROPERÚ received a special order to manage the south west concession for three years (until December 2026) by SD N° 019-2023-EM at the end of 2023 [62]. Furthermore, it is important to note that in the NG distribution concession in Tumbes, the NG of the Tumbes concessionaire was unable to carry out the commercial commissioning on the date specified in the concession contract (06/29/2021), resulting in a breach of contract and an arbitration claim [63]. In Peru, the main regulations on NG distribution activities are as follows: SD N° 040-2008-EM, the single ordered text of the NG distribution regulations through duct network that collects the regulations for the activity of the public service of the NG distribution through duct network [64], and SD N° 058-2010-EM, which approves the regulation law of the creation of

municipal companies in charge of providing the public service of NG supply through duct network in the scope of district and provincial municipalities (Law N° 29496), which applies in areas where the State has not granted permission for the distribution of NG at home [65].

The *commercialization* activities include storing, packaging, transporting, and dispensing the finished NG products [66]. In Peru, NG is commercialized by properly authorized companies that import, export, store, transport, distribute, or sell VNG, LNG, and CNG [67]. In Peru, the main regulations on NG commercialization activities are: SD N° 057-2008-EM, which approves the regulations for the commercialization of CNG and LNG [68]; SD N° 006-2005-EM, which approves the regulations for the installation and operation of establishments for the public sale of VNG [69]; and SD N° 009-2006-EM, which declares the use of VNG of national interest and modifies the regulation for the installation and operation of establishments for the public sale of VNG [70].

3.2.3. Demand in domestic and foreign natural gas markets

NG produced in Peru is highly demanded in domestic and international markets. The following sectors drive demand for NG in Peru's domestic market: EPG, industrial consumers, VNG, residential, and commercial. The demand for NG on the foreign market is made up of exports to international markets (the main destination markets for Peruvian LNG exports are Canada and Mexico in North America, UK and Spain in Europe, and China, Japan, Taiwan, Thailand and South Korea in Asia) [1,56,60]. Regarding total NGC in the internal and external markets, the consumption of the EPG sector increased from 419 MMcfd in 2021 to 484 MMcfd in 2022 (a variation of 16 %), the consumption corresponding to the industrial sector increased from 161 MMcfd in 2021 to 164 MMcfd in 2022 (an increase of 2 %), NGC of the vehicle sector increased from 53 MMcfd in 2021 to 66 MMcfd in 2022 (a change of 25 %), consumption of the residential and commercial sector increased from 24 MMcfd in 2021 to 29 MMcfd in 2022 (an increase of 21 %), and consumption corresponding to the external sector (exports) increased from 341 MMcfd in 2021 to 460 MMcfd in 2022 (a change of 35 %) [56].

In 2010, Peru's EPG was 32111.58 GWh. In this year, the EM of the Peruvian electricity sector was composed of hydro 59.1 % (18964.56 GWh), NG 35.6 % (11446.53 GWh), wind 0.0 % (0.0 GWh), solar 0.0 % (0.0 GWh), coal 3.3 % (1066.92 GWh), bagasse 0.2 % (77.48 GWh), residual 1.7 % (556.09 GWh), diesel 0.0 % (0.0 GWh), and biogas 0.0 % (0.0 GWh). In 2023, the EPG totaled 58465.56 GWh. This year, the Peruvian EPG by ES was composed of hydro 47.7 % (27901.29 GWh), NG 44.6 % (26089.50 GWh), wind 4.1 % (2368.03 GWh), solar 1.7 % (978.51 GWh), coal 0.0 % (0.0 GWh), bagasse 0.5 % (272.03 GWh), residual 0.04 % (26.14 GWh), diesel 1.3 % (760.60 GWh), and biogas 0.1 % (69.46 GWh) [71]. As can be seen, in 2023, NG, slightly below water, will be the second most important ES in Peru's EPG. Similarly, between 2010 and 2023, the share of NG in the EPG increased from 35.6 % to 44.6 %, the share of RS, such as wind and solar, increased from 0.0 % to 5.8 %, the share of diesel increased from 0.0 % to 1.3 %, and the shares of hydro, coal, and residual decreased from 59.1 % to 47.7 %, 3.3 %–0.0 %, and 1.7 %–0.04 %, respectively.

In terms of Peruvian LNG exports, the value was 1552.8 (US\$MM FOB) in 2022 and 3166.2 (US\$MM FOB) in 2023 [12]. These figures represented 4.8 % of total Peruvian exports by economic activity group in 2022 and 2.3 % in 2023, correspondingly.

3.2.4. Impacts of natural gas industry

The NG industry's activities have impacted the Peruvian economy (boosting GDP, generating employment, producing improvements in the trade balance, reducing CO₂ emissions) and resources for the Peruvian state through taxes and royalties. These are the results. Similarly, investments in NG exploration (October 2023: US\$0.23 million; [72]) and exploitation (October 2023: US\$22.65 million; [72]) generate additional resources for regional and local governments via the NG canon. The NG industry has also impacted the industrial, commercial,

⁴ The amount of the DNG that is not transported by PGT is reinjected into Lot 88 field [1].

residential, electrical, and transportation sectors [1].

According to Macroconsult, a Peruvian consulting firm, from 2004 to 2019, the average annual direct contribution to the Peruvian economy from the exploitation of the Camisea NG was 1.1 % of GDP, with the hydrocarbon sector contributing 39 % of GDP [11]. According to the Peruvian Hydrocarbons Society (PHS) and Cálidda, NG contributes more than 10493 million PEN to the country's GDP and has created over 55,000 direct and indirect jobs in 2022 alone [73]. Similarly, Macroconsult reports that the Camisea NG project created 30000 direct and indirect jobs per year between 2004 and 2019 [74], while also saving the country US\$100,000 million [11]. According to Ref. [1], during this period, tax revenues generated by the NG industry -primarily through income tax, royalties, and the canon-reached 27000 million US dollars capitalized by 2019, with NG royalties accounting for 79 %. Until 2004, the Peruvian State's collection of NG royalties accounted for only 4 % of total hydrocarbon royalties [1]. By the end of 2022, those for NG had 53.26 %. In 2022, the Peruvian State collected 1843.08 MMUS dollars in hydrocarbon royalties [66].

The gas canon is the quota that Peruvian local and regional governments have on the income received by the state from income taxes and royalties derived from the exploitation of NG and NGL [55]. During January to December 2023, the NG canon transferred was 3321.5 million PEN, which was 8.7 % less than the amount transferred during the same period in 2022. In December 2023, transfers per NG canon were 279.4 million PEN, 10.7 % higher than in December 2022 [72]. In January 2024, the canon for NG was 18.9 % lower than in January 2023 (PEN 330.7 million), and 4.1 % lower than in December 2023 [75].

The cumulative impact on hydrocarbon trade balance (HTB) has increased by 42 % compared to the results from 2013. As a result, the value of the benefit of NG industry development in the HTB reached 55083 million US dollars capitalized in 2019 ([1], p. 133).

It is estimated that the Camisea Project's CO₂ emissions reductions in the electricity, industry, and vehicular transportation sectors will total 72 million tCO₂ and cost 2296 million US dollars between 2004 and 2019 ([1], p. 153).

Finally, in terms of the impacts of the NG industry on the industrial, commercial, residential, electrical generation and vehicular sectors, according to Ref. [1], the accumulated savings during the period 2004–2019 in the industrial sector were equal to US\$10,089 million in Lima updated to 2019, in the residential sector of Lima and Callao the net savings were more than US\$740 million updated to 2019. The net savings in the EPG sector were equal to US\$3081 million updated to 2019, in the commercial sector the net savings accumulated in the same period reached US\$54 million, while according to PHS and Cálidda, VNG users, during the period 2004–2023, have saved more than 61000 million PEN [73].

4. Material and methods

4.1. Materials

Data collection and processing was carried out with Microsoft Excel and econometric estimates were performed with EViews software (version 13).

4.2. Variables and data

The variables under study in this article were real GDP measured in constant 2007 million PEN, NG measured in m³, and real Tr (the sum of real imports and real exports of goods and services) measured in constant 2007 million PEN. Central Reserve Bank of Peru (BCRP) and Energy Institute (EI) provided us with annual data on our research variables for the years 1965–2022. The data on GDP, exports, and imports were collected from BCRP [76–78] and the data on NGC was obtained from EI [54]. The original NGC data are in billions of m³ and exclude NG converted to liquid fuels, but they do include coal

derivatives and NG used in the gas-to-liquids conversion. In this study, the original NGC data were converted into m³.

The sample size for this study is 58. From 1965 to 2022, the average value of real GDP was 2.52410E⁺⁰⁵ (constant 2007 million PEN), the average value of Tr was 1.12822E⁺⁰⁵ (constant 2007 million PEN), and the average value of NGC was 2.22488E⁺⁰⁹ m³. Time series GDP_t , Tr_t , and NGC_t presented very high standard deviations: 1.40742E⁺⁰⁵ million PEN, 8.42559E⁺⁰⁴ million PEN, and 2.95874E⁺⁰⁹ m³, respectively. To reduce their variability, we transformed all variables into natural logarithms. Lastly, Table 1 shows strong positive correlations between GDP_t and NGC_t , GDP_t and Tr_t , and NGC_t and Tr_t .

Fig. 1 shows real GDP growth periods from 1965 to 2022 as follows: 1965–1977, 1978–1981, 1983–1987, 1993–1997, and 1998–2019. Similarly, between 1981 and 1983, as well as 1987 and 1990, real GDP experienced declines. Between 1965 and 1999, the NGC oscillated around 4.82556E⁺⁰⁸ m³, with a peak growth period between 2000 and 2012. Furthermore, from 2013 to 2019, the NGC oscillated around 7.71417E⁺⁰⁹ m³. In 2020, both series observed the coronavirus epidemic's negative effects. Fig. 1 also shows that Camisea Gas, Peru's most important NG reserve, was exploited in 2004. Similarly, this figure shows that Peru began exporting LNG in 2010 when Peru LNG began operations at the Pampa Melchorita liquefaction plant, which has a capacity of 4.45 million tons per year [79].

Fig. 2 shows that from 1965 to 1990, the series Tr_t oscillated around 4.66918E⁺⁰⁵ million PEN. Similarly, this series showed an increasing trend from 1991 to 2019. Again, in 2020, the coronavirus's negative effect is observed. Likewise, this figure depicts the start of Camisea gas extraction and operations at the Pampa Melchorita plant in 2004 and 2010, respectively.

4.3. Theoretical framework⁵

The generic and specific functional forms of the theoretical model used in this article, based on [22], state that EG (measured by GDP) is dependent on NGC and Tr. The generic model is characterized by:

$$GDP = f(NGC, Tr) \quad (1)$$

According to GH hypothesis, energy consumption (NGC) contributes positively to EG Ref. [29]. Likewise, in accordance with the trade-led economic growth (TLE) hypothesis an increase of Tr leads to a greater EG Ref. [30].

The specific functional form –a logarithmic linear model– is given by the following:

$$\ln GDP = a + b \ln NGC + c \ln Tr \quad (2)$$

Where a is a constant, and b and c are the partial elasticities of GDP with respect to NGC and with respect to Tr, respectively. Based on [22], it is

Table 1
Correlation matrix (variables in levels).

	GDP_t	NGC_t	Tr_t
GDP_t	1.00000		
NGC_t	0.96127*	1.00000	
Tr_t	0.98850*	0.94579*	1.00000

Compiled by the authors. (*) indicates a 0.05 significance level.

⁵ In the literature that analyze the GDP-NGC nexus, various authors use models with a neoclassical production function that considers capital, labor, NGC, and other control variables –trade openness, exports, foreign direct investment, technological progress, EC from coal and oil, among other– as determinants of GDP. Readers interested in this type of theoretical framework may, for example, refer to Refs. [23,25,36,52], [27,30,31], [80].

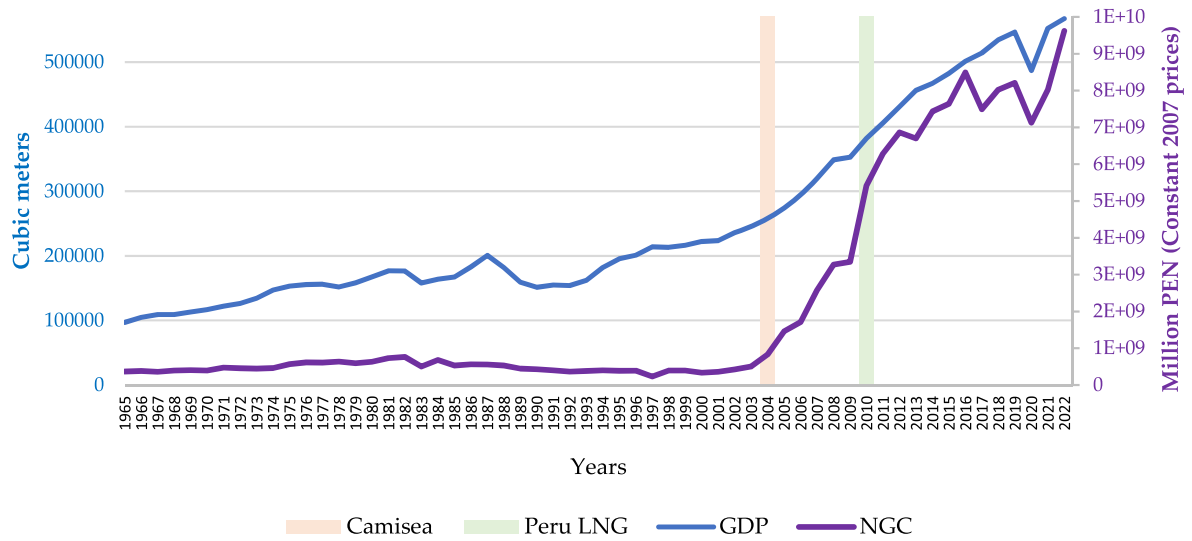


Fig. 1. Evolution of real GDP and natural gas consumption in Peru from 1965 to 2022 (compiled by the authors).

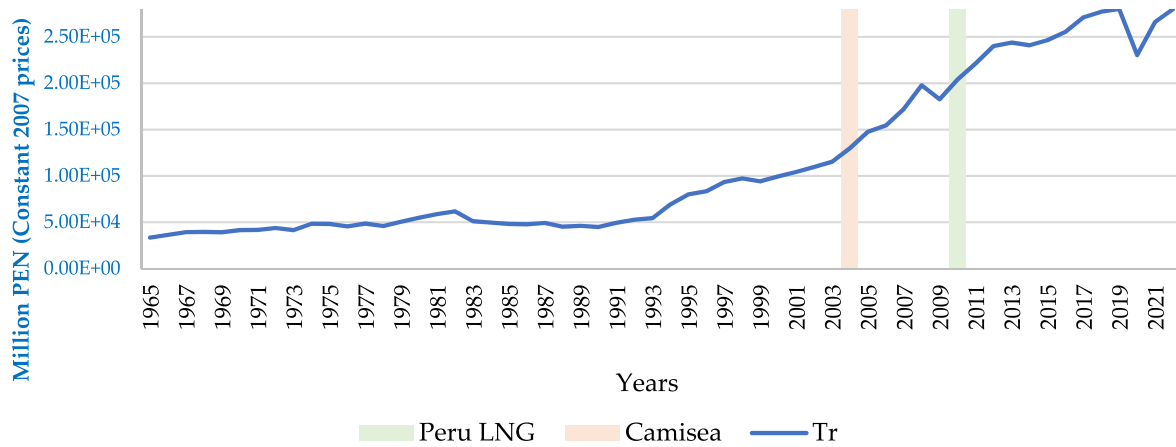


Fig. 2. Evolution of real trade Balance in Peru from 1965 to 2022 (compiled by the authors).

expected that $b > 0$ and $c > 0$. Thus, an increase/decrease in NGC, ceteris paribus, is expected to produce an increase/decrease in GDP. Analogously, an increase/decrease in Tr is expected to generate an increase/fall in GDP.

4.4. Methodology

The methodology used in this article is the same as in Refs. [81,82], and it consists of the stages listed below: (i) Stationarity analysis (order of integration) of all logarithmic series. For this purpose, the ADF is used for identify the presents of the unit roots on the log-series. (ii) Analysis of causality using stationary log series (If necessary, the first differences operator is applied to each log series until stationarity is confirmed using the ADF test). To this end, the pairwise GC test is used. The Akaike information criteria (AIC), Hannan–Quinn information criteria (HQ), and Schwarz information criteria (SIC), determine the optimal lag for applying the Granger test. If all series show feedback causality, then the multiple-equation model is preferred, for example the VAR model. (iii) After verifying that the log series has an order of integration lower than $2-I(0)$ or $I(1)$ –, we estimated the following ARDL model:

$$ARDL(p, q, r) : Y_t = \alpha + \sum_{k=1}^p \omega_k Y_{t-k} + \sum_{k=0}^q \sigma_k X_{t-k} + \sum_{k=0}^r \vartheta_k Z_{t-k} + \mu_t \quad (3)$$

(iv) To examine the LR equilibrium relationship between all log

series, the ARDL bounds test for cointegration is used in the estimated $ARDL(p, q, r)$ model. (v) After assessing the cointegration of all log series, a cointegrating relation and an ECM are estimated to determine the short- and long-run elasticities. (vi) Finally, the estimated ECM undergoes the following diagnostic tests: serial correlation, heteroskedasticity, normality, residual stationarity, functional form, and structural stability.

[83] shown another cointegration method, the Engle-Granger cointegration test [84], when the level time series—non-differenced series—are non-stationary, $I(1)$, they could maintain a long-run equilibrium relationship. Starting with the following model:

$$Y_t = \beta_1 + \beta_2 X_t + \beta_3 Z_t + u_t \quad (4)$$

If the residuals $\hat{u}_t$ obtained from the estimation follow a stationary process, the series Y_t, X_t , and Z_t are cointegrated. Therefore, if the null hypothesis of no cointegration between the series is rejected, then the slope coefficients are long-run elasticities, and the slope coefficients obtained in the ECM model—where all the log-series are differenced—are short-run elasticities.

5. Results

The next section presents the results of the econometric estimates and diagnostic hypothesis tests performed in this paper.

5.1. Stationarity test

Table 2 depicts the outcomes of ADF URT. Based on these findings, it is deduced that all variables are nonstationary in levels (i.e., all log series are first-order integrated processes, $I(1)$), but in first differences, all variables are stationary (That is, the first differences of all log series are integrated processes of zero-order, $I(0)$).

5.2. Pairwise Granger causality

Table 3 shows a causal relationship of neutrality between $\Delta \ln NGC_t$ and $\Delta \ln GDP_t$, one bidirectional causality relationship between $\Delta \ln Tr_t$ and $\Delta \ln GDP_t$, and one relationship of neutrality among $\Delta \ln Tr_t$ and $\Delta \ln NGC_t$. These results allow us to conclude that GC validated NH in Peru. Also, the presence of one feedback causality after testing the six null hypotheses suggested avoiding the VAR model.

5.3. Estimated autoregressive distributed lag model and cointegration test

The three $I(1)$ Time series allow us to estimate the $ARDL(1, 0, 1)$ model, with restricted constant and no trend, in which $\ln GDP_t$ is the regressed variable and $\ln NGDP_{t-1}$, $\ln GNC_t$, $\ln Tr_t$, $\ln Tr_{t-1}$, and two dummies variables (considered to reach the structural stability) are the regressors. The optimal lag lengths were found using the AIC, with a maximum of four lags in the regressors and dependent variable.

After applying the ARDL bounds test with restricted constant and no trend to the estimated $ARDL(1, 0, 1)$ model the F -statistic value was 11.33107, and as all $I(0)$ stationary bounds and all nonstationary bounds $I(1)$ —regardless of the level of statistical significance (1, 5, or 10 %) and for sample sizes greater than or equal to 55—fall below this value, the hypothesis of no cointegration among the three log series was not accepted at the 0.05 level.

5.4. Estimated cointegrating relation

Table 4 shows the estimated cointegrating relation. In this model, $\ln NGC_t$ and the first-order lag of $\ln Tr_t$ were considered as regressors of $\ln GDP_t$. In the LR, the impact of $\ln NGC_t$ on $\ln GDP_t$ is 0.10022, and the impact of $\ln Tr_{t-1}$ on $\ln GDP_t$ is 0.58990—both at the 0.05 significance level—. These results permit us to conclude that NH is validated by cointegration in Peru.

5.5. Estimated error correction model

The verification of the cointegration between $\ln GDP_t$, $\ln NGC_t$, and $\ln Tr_t$ under the ARDL approach led to the estimation of the corresponding ECM (refer to Table 5). To achieve structural stability in 1986 and 1987, this model used the first difference of $\ln Tr_t$ as a regressor, along with two dummy variables. Table 5 also shows a significant negative error correction term (ECT) in the ECM, which supports the results of the LR equilibrium nexus for all research variables. A positive and significant effect of $\Delta \ln Tr_t$ is found on $\Delta \ln GDP_t$. The estimated ECM has an adequate adjustment with an R^2 -adjusted value above 0.5. Diagnostic tests validate that the errors are normally distributed

Table 2
Augmented Dickey-Fuller test.

Variables	Auxiliary model	Lag***	Probability	Integration Order*
$\ln NGC_t$	None**	0	0.99326	$I(1)$
$\Delta \ln NGC_t$	None**	2	0.02864	$I(0)$
$\ln Tr_t$	None**	0	0.99990	$I(1)$
$\Delta \ln Tr_t$	Constant	0	0.00000	$I(0)$
$\ln GDP_t$	None**	0	0.99999	$I(1)$
$\Delta \ln GDP_t$	Constant	0	0.00001	$I(0)$

Compiled by the authors. (*) indicates a 0.05 significance level. (**) denotes series without constant or trend. (***) Schwarz information criteria.

Table 3
Granger causality test.

Causality	Lag	Criterion	Probability
$\Delta \ln NGC_t \rightarrow \Delta \ln GDP_t$	1	SIC	0.58450
$\Delta \ln GDP_t \rightarrow \Delta \ln NGC_t$	1	SIC	0.25870
$\Delta \ln Tr_t \rightarrow \Delta \ln GDP_t$	2	AIC	0.02532*
$\Delta \ln GDP_t \rightarrow \Delta \ln Tr_t$	2	AIC	0.05220**
$\Delta \ln Tr_t \rightarrow \Delta \ln NGC_t$	1	SIC	0.41208
$\Delta \ln NGC_t \rightarrow \Delta \ln Tr_t$	1	SIC	0.48989

Compiled by the authors. (*) denotes a 0.05 significance level. (**) represents a 0.1 significance level.

Table 4
Estimated cointegrating relation*.

Variables	Coefficients
$\ln NGC_t$	0.10022**
$\ln Tr_{t-1}$	0.58990**
Constant	3.55691**
Regressand variable	$\ln GDP_t$
Number of coefficients	3
Number of included observations	57

Compiled by the authors. (*) indicates restricted constant and no trend. (**) denotes a 0.05 significance level.

Table 5
Estimated error correction model*.

Variables	Coefficients
ECT	-0.23025**
$\Delta \ln Tr_t$	0.44184**
Dummy ₁₉₈₆	0.11473**
Dummy ₁₉₈₇	0.12194**
Regressand variable	$\Delta \ln GDP_t$
Number of coefficients	4
Number of included observations	57
R^2 -adjusted	0.74684
Diagnostic tests	Probability
Breusch-Godfrey***	0.45171
White non-cross terms test	0.18045
White with cross terms	0.27406
Breusch-Pagan-Godfrey	0.17661
Glejser	0.21983
Augmented Dickey-Fuller	0.00000
Jarque-Bera	0.07229
Ramsey-Reset	0.95026

Compiled by the authors. (*) indicates restricted constant and no trend. (**) denotes a 0.05 significance level. (***) indicates that AIC, SIC, and HQ were taken into account for the first-order lag.

(verified by the Jarque–Bera test) that heteroskedasticity is not detected (verified by the White, Breusch–Pagan–Godfrey, and Glejser tests), and that there is no serial correlation (verified by the Breusch–Godfrey test) in the ECM. The Ramsey RESET test validates the ECM's correct specification. The Dickey–Fuller test indicates that the residuals are $I(0)$ process at the 0.05 level. Finally, the CUSUM and CUSUMSQ tests provide evidence for structural stability (Fig. 3).

5.6. Alternative cointegration method

We applied the Engle–Granger cointegration test as a second method to contrast the findings obtained by the ARDL cointegration method. Considering the dependent variable $\ln GDP_t$, and the independent variables $\ln Tr_t$ and $\ln NGC_t$ with constant in the linear regression model, the τ -statistic of the Engle–Granger cointegration test is equal to -2.34920 with p -value equal to 0.5718, concluding the absence of the cointegration between the non-stationary log-variables $I(1)$, at 5 % significance level.

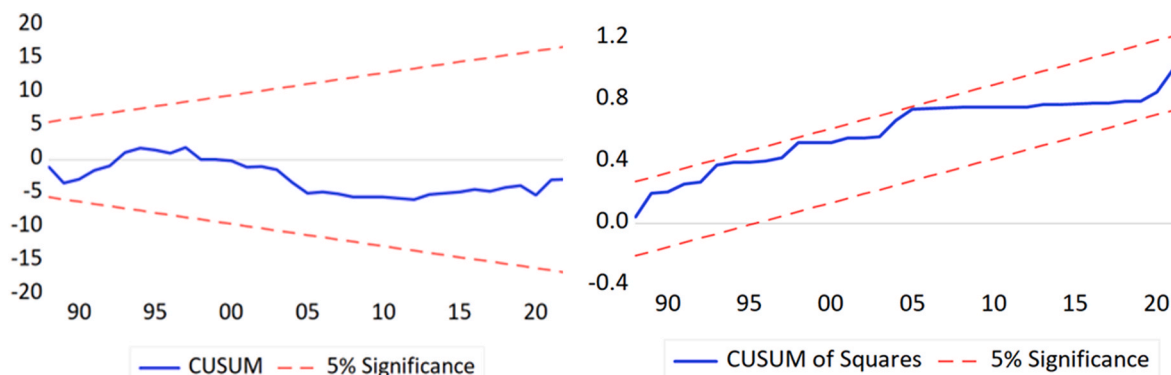


Fig. 3. Cusum and Cusum of squares of recursive residuals.

6. Discussion

First, we found that at the 5 % level of significance, the first differences in the log series NGC, Tr, and GDP are stationary. Likewise, the pairwise GC test supported NH between NGC and EG in Peru for the sample period at a 5 % significance level. [24,29], along with us, confirmed NH in South Korea and Turkey, respectively. However, contrary to our results, [22] confirmed GH among GDP and NGC in Saudi Arabia using the TY GC test.

Based on the literature review elaborated in this article, the following four hypotheses about the NGC–EG nexus were supported through diverse causality tests:

- *Findings on the GH hypothesis:* Unlike our results [22–24,29], and [27, 31] validated the GH in Pakistan (1972–2010), China and Thailand (1980–2012), Saudi Arabia (1968–2016), Turkey (1996–2010 and 1997–2011), Pakistan (1977–2013), and Malaysia (1980–2010), respectively. TY GC supports the results obtained by Refs. [22,24]. The evidence reported by Ref. [23] is based on the IAA GC approach. Hacker and Hatemi-J causality validate the results found by Ref. [29]. Lastly, the findings uncovered by Refs. [27,31] are verified by VECM GC approach.
- *Findings about the CH hypothesis:* In contrast to our results [24,26,32], and [33,34] confirmed the CH in Bangladesh, Singapore, and Malaysia (1980–2012), Malaysia (1980–2014), Malaysia (1971–2012), Japan (1980–2020), and Pakistan (1980–2010), respectively. These five authors obtained their results using the VECM GC approach, the TY GC test, the GC test, TY GC, and the IAA GC technique, respectively.
- *Findings about the FH hypothesis:* In contrast to our results [21,24,25, 27,29,30], [32,35,36], and [13] validated the CH in Germany and Japan (1970–2018), Japan, India, and Pakistan (1980–2012), France (1990–2014), Malaysia (1980–2010), Turkey (2001–2014), Indonesia (1980–2017), Malaysia (1971–2012), Nigeria (1981–2015), Turkey (1995–2012), and Turkey (1988–2021), respectively. The results of [21,24,29,30,35], and [13] were supported by FGC and MS-GC (in crisis and growth regimens), causal direction from dependency (D2C) algorithm with Machine Learning experiments, the causality of Hacker & Hatemi-J, the variance decomposition method, asymmetric nonlinear causality of Hatemi-J, and the TY GC test, respectively. Lastly, all outcomes obtained by Refs. [25,27,32], and [36] were supported by VECM GC approach.
- *Findings about the NH hypothesis:* Consistent with our results, [24,29], supported NH in South Korea from 1980 to 2022 and in Turkey from 1983 to 2017. However, while we validated NH in Peru using the pairwise GC test, [24], and [29] validated it using the Hacker & Hatemi-J causality test and the TY GC test, respectively.

Second, the ARDL CT with GDP as the dependent variable verifies the

GH in Peru, where the LR elasticity of GDP_t in relation to NGC_t is 0.10022 at the 5 % level of statistical significance. Our cointegration results reveal that the SR elasticity of GDP_t with respect to Tr_t is 0.44184, and that the LR elasticity of GDP_t with regard to Tr_{t-1} is 0.58990, at 5 % level of significance. These results discussed above are justified with the ARDL cointegration test, because we do not find cointegration by Engle-Granger method. As stipulated by Ref. [25], the ARDL cointegration test resolves the lack of capacity of the Engle-Granger method to test hypotheses about the estimated coefficients in the LR, and presents better properties respect to other multivariate cointegration methods in small samples [85].

The literature review of this paper shows that only GH about the NGC–EG nexus has been validated through diverse cointegration methods. The findings, by cointegration, about the LR effects of NGC in relation to EG are showed in Refs. [22,23,25–28,30,31] confirming GH in France (1990–2014), Pakistan (1972–2010), Malaysia (1980–2014), Indonesia (1980–2017), Malaysia (1982–2020), Saudi Arabia (1968–2016), Pakistan (1977–2013), and Malaysia (1980–2010), respectively. However, although in Peru we validated GH in the LR with the ARDL bounds test [23,25–28], and [30] also validated GH with the ARDL bounds test, but not only in the LR but also in the SR. [22], just as we, verified GH with the ARDL bounds test in LR. Finally, [31] confirmed GH with the Johansen cointegration method in LR. The positive and significant LR elasticities of GDP_t with respect to NGC_t estimated by these eight authors were 0.4115, 0.2016, 0.028, 0.453, 0.298, 0.157, 0.2623, and 0.0239, respectively.

Third, between 1965 and 2022, we confirmed a bidirectional GC between Tr and real GDP in Peru at the 10 % statistical significance level. Analogous to our results, [38,39,46–48] found a bidirectional causality between Tr and EG in OECD countries, GCC countries, Asia-Pacific countries, Australia, and in forty-six emerging market and developing economies, correspondingly. Contrary to our results, using a VECM GC approach, [27] found a one-way causality association running from Tr to GDP in Malaysia, in both S&LR. Divergent to our results, [22, 33,43] validated a neutral causality between Tr and GDP in Saudi Arabia, Japan, and in Brazil and Russia, respectively. Using the other methodology, the findings on the SR effects of Tr with respect to EG are reported: [28,33], estimate positive and significant SR elasticities of GDP_t with respect to Tr_t equals to 0.1716 and 0.030644 in Malaysia and Japan, respectively.

Finally, we confirmed a neutral GC between real Tr and NGC in Peru at the 10 % significance level. Using the TY GC test, [22] validated a neutrality causality between Tr and NGC in Saudi Arabia, in concordance with our results. Similar to our results, using the Emirmahmutoglu and Kose heterogeneous mixed panel causality test, [40] supported neutrality between NGC and Tr in selected sub-Saharan African countries. Contrary to our results, using a VECM GC approach, [27] found a one-way causality association that runs from Tr to NGC in Malaysia, both in S&LR. Divergent to our results, employing panel VECM GC test

for panel, [38,39] validated a neutral causality between NGC and Tr in twenty six OECD countries, and in GCC countries, respectively. Unlike us, using Dumitrescu and Hurlin panel causality test, [37] found a bidirectional causality between NGC and Tr in fifteen developing countries with high levels of NG consumption.

7. Conclusions: policy implications and future research

This study analyzes the dynamic NGC–EG nexus in Peru from 1965 to 2022 considering Tr as non-omitted variable. From the results obtained in this research, the following conclusions are derived: First, the GC test results in pairs validated the NH between NGC and EG, allowing us to conclude that policies to conserve or increase NGC will have no effect on Peru's EG and vice versa over the sample period. Besides, the pairwise GC test confirmed a bidirectional relationship between Tr and GDP (validating both the TLE and ELT hypotheses). Second, there is cointegration in all variables, it is noted that Tr and NGC are drivers of EG in Peru in the LR, verifying that NGC has a positive impact on EG in the LR, while for Tr results showed that, in the LR, the value of Tr from the previous year has a positive effect on the current value of GDP; and that in the SR, the current value of Tr has a positive effect on GDP. Last, the pairwise GC test also validated a neutrality causal relationship between NGC and Tr, allowing us to conclude that policies to preserve or increase NGC will not affect Peru's Tr or vice versa during the study period.

In view of these results, the following policy implications arise: First of all, although the causal relationship between NGC and EG shows evidence in favor of the neutrality hypothesis (NH), given the presence of cointegration between all variables (with GDP as a dependent variable) in the LR, we can conclude that an increase or decrease in NGC, *ceteris paribus*, will positively or negatively impact EG, respectively. Therefore, implementing CO₂ reduction policies aimed at reducing NGC will negatively affect Peru's EG in the LR. In that regard, we recommend that Peruvian policymakers apply this type of policy to most PFF, such as oil and diesel, rather than NG, whose CO₂ emissions are much lower –and whose economic impacts during the period 2004–2019 have already been reported– and gradually introduce non-polluting RE such as green hydrogen, solar, and wind to the Peruvian EM. It should be noted that this policy recommendation constitutes evidence that strengthens the importance of the energy transition objective for Peru, which is part of its National Energy Policy 2010–2040, approved by Supreme Decree (SD) No. 064-2010-EM [15]. Thus, the NGC can be constituted as an alternative energy source to fossil fuels, its impact with respect to the latter also being positive on economic growth, but less in terms of environmental degradation given the lower amount of CO₂ emissions that are derived of the same.

On the other hand, given the positive impact that NGC has on EG in the LR, for the purposes of continuing to promote the NGC industry in the country, measures such as those highlighted by Ref. [57]: (i) Regulatory adjustments according to the maturity of the NG industry, (ii) the development of the petrochemical industry, (iii) The future exploitation of shale gas, (iv) NG supply authorizations, (v) National gas ducts system, among others. Additionally, with relation to the evidence in favor reported for the TLE hypothesis verified, we first recommend that policymakers strengthen trade liberalization policies to ensure Peru's LR EG. According to Ref. [86], three major policies stand out: (i) productive diversification with high-value-added products; (ii) exchange rate policy; and (iii) tariff and non-tariff policy. Thus, concerning the first policy, it is appropriate to direct international trade (IT) toward new value-added sectors. Following to Ref. [86], Peru has maintained the same export structure (raw materials) for centuries, in contrast to economies such as South Korea, China, and Japan, which rapidly changed from raw materials to products with higher technological content.

Regarding the second policy, as previously stated [86], exchange rate policy is a relevant tool of industrial policy. Exchange rate appreciation can undermine all attempts to vary a country's export base. On

the contrary, an appropriate exchange rate policy can be useful in combating industrial opening, avoiding unfair competition, and promoting sectoral productive development. Finally, regarding tariff and non-tariff policy, it is considered important to establish appropriate standards that do not discourage commercial exchange; the more each country specializes in its comparative advantage, the fewer incentives there are to impose restrictions that limit the advancement of the IT. On the contrary, greater integration based on appropriate standards encourages greater dynamism in IT, resulting in increased EG and job creation, business activity, and a reduction in poverty and income inequality.

For its part, with respect to the evidence a favor of the ELT hypothesis, using the gravitational model of IT as a reference, IT between trading partners is explained by the GDP of the country in question or the issuer of IT [87,88], it is recommended to preserve the country's dynamism and economic stability, which in Peru is in charge of the BCRP and the Ministry of Economy and Finance. According to the gravity model, the GDP of the IT-emitting country acts as a force of attraction for IT, because the higher income or economic size of the exporting country is associated with greater use of its resources, as well as the development of economies of scale in its production, which encourages further development of IT. As a result, the importance of permanent EG in promoting increased IT flows. Likewise, the evidence in favor of the hypothesis of neutrality between NGC and Tr, substantiates the relevance of both variables individually for the EG and at the same time strengthens the importance of the policy implications previously raised for each of them in order to boost the economic activity of the country.

Finally, we propose as future lines of research: (i) Estimate the impact of NGC on CO₂ emissions, as showed in Refs. [53,89–91], (ii) Analyze how the use of NG, as a substitute for fossil fuels with higher levels of polluting emissions, would affect the EG, as exposed in Ref. [52], (iii) Estimate a non-linear dynamic association between NGC, EG, and other variables (see for example, [35,92]), (iv) Examine the relationship between EG and energy sources other than NG, considering non-energy variables (see for instance, Ref. [93]), (v) Estimate the relation between EG and CO₂ emissions, considering renewable and non-renewable energy variables, and other variables– (see for example, [94–103]).

CRedit authorship contribution statement

Ciro Bazán: Writing – original draft, Writing – review & editing, Conceptualization, Investigation, Validation, Methodology, Project administration. **Víctor Álvarez-Quiroz:** Methodology, Data curation, Validation, Writing – review & editing, Visualization, Supervision. **Juan Morocho:** Methodology, Data curation, Validation, Writing – review & editing, Visualization, Supervision. **Juan Castillo:** Methodology, Data curation, Validation, Writing – review & editing, Visualization, Supervision. **Rolando Herrera:** Methodology, Data curation, Validation, Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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